

Operational Manual: Transition Matrix Extraction and Macroscopic Metasurface Scattering

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Abstract

This manual outlines the execution protocol and physical theory for the computational framework designed to extract the transition matrix of isolated metamaterial unit cells and model the macroscopic scattering response of large-scale metasurfaces [1, 2]. It delineates the relationship between the finite-element near-field extraction, algebraic transition matrix projection, and macroscopic multiple-scattering aggregation.

1 System Architecture and Input/Output Summary

The workflow is strictly partitioned into three sequential stages: finite-element near-field extraction, algebraic transition matrix projection, and macroscopic multiple-scattering aggregation. The transition matrix describes the waves scattered by the localized geometry and must not be conflated with a transfer matrix.

Table 1: Computational Stage I/O Summary

Computational Stage	Script Name	Primary Inputs	Primary Outputs
1. Full-Wave Extraction	<code>extract_cst_near_fields_2.py</code>	CST Template, w, h, f_{max}	Cartesian fields, <code>excitation_metadata.json</code>
2. T-Matrix Projection	<code>compute_t_matrix_projection_2.py</code>	<code>metadata.json</code> , Fields, L_{max} , $r_{monitor}$	Isolated T-Matrix Operator (<code>T_Matrix.npy</code>)
3. Array Aggregation	(Planned Architecture)	Array of <code>T_Matrix.npy</code> , Translations	Macroscopic S-Parameters (S_{11}, S_{21})

2 Operational Protocol and Execution Setup

To execute the automated pipeline, a standalone Python environment must be configured with the CST Studio Suite libraries. This is achieved via the following console command: `pip install --no-index --find-links "<CST_STUDIO_SUITE_FOLDER>/Library/Python/repo/simple" cst-studio-suite-link`.

2.1 Stage 1: Full-Wave Extraction (`extract_cst_near_fields_2.py`)

This stage automates the full-wave computational electromagnetic extraction utilizing the Frequency Domain Solver within CST Microwave Studio.

Physics Setup and Constraints:

- **Substrate Omission:** The computational CAD model within CST Microwave Studio must strictly omit the dielectric substrate. The primary full-wave simulation must evaluate the metallic or dielectric resonator in complete isolation. Including the substrate within the computational domain violates the spherical symmetry requisite for the exact projection of the scattered near-fields onto the free-space vector spherical wave functions (VSWFs).
- **Boundary Conditions:** The unit cell must be simulated strictly in isolation. The enforcement of periodic boundary conditions (Floquet modes) is mathematically erroneous

for extracting a localized multipole response. Perfectly Matched Layers (PMLs), designated as "Open (add space)", must be applied on all six sides of the bounding box to allow scattered fields to radiate freely in 4π steradians.

- **Spherical Field Monitor Placement:** To project the simulated fields onto the orthogonal VSWF basis, a discrete spherical monitor surface, S , must be defined. This sphere must be positioned exclusively within the bounding box and inside the PML boundaries. To prevent intersection with the physical scatterer while simultaneously avoiding non-physical absorption artifacts from the PML, the radius of the monitor, $r_{monitor}$, is mathematically constrained to $r_{scatterer} < r_{monitor} < r_{PML}$.

Execution Routine: The script generates a discrete spectrum of incident plane waves distributed across the solid angle utilizing a rigorous Lebedev quadrature grid. This guarantees exact numerical integration for the spherical harmonic basis and establishes a highly overdetermined system to mitigate spatial meshing noise. It iteratively computes the response and exports the complex electric and magnetic Cartesian near-fields alongside an `excitation_metadata.json` manifest.

2.2 Stage 2: Transition Matrix Projection (`compute_t_matrix_projection_2.py`)

This script executes the linear algebra necessary to project the exported near-fields onto the orthogonal VSWF basis.

Physics Setup and Constraints:

- **Numerical Integration:** Cartesian trilinear interpolators evaluate the complex fields on the discrete spherical monitor surface. The script utilizes a Gauss-Legendre quadrature grid for numerical surface integration to evaluate the expansion coefficients.
- **Substrate Reintroduction:** The substrate, omitted in Stage 1, is systematically reintroduced during this algebraic phase using rigorous analytical matrix operations.
 - *Infinite Half-Space Substrate:* If the array is situated on a semi-infinite substrate, a reflection matrix $\mathbf{R}$ is defined by expanding outgoing VSWFs into a continuous plane-wave spectrum via the Weyl identity, reflecting them via standard Fresnel coefficients, and re-projecting onto the incoming VSWF basis. The coupled response is $\mathbf{T}_{coupled} = [\mathbf{I} - \mathbf{T}_{iso}\mathbf{R}]^{-1}\mathbf{T}_{iso}$.
 - *Finite Substrate Pad:* If the substrate is a truncated volumetric pad, it is treated algebraically as a secondary distinct scatterer ($\mathbf{T}_{sub}$). Employing the Translation Addition Theorems ($\mathbf{A}^{12}, \mathbf{A}^{21}$), the Superposition T-matrix Method yields $\mathbf{T}_{meta}^{(c)} = [\mathbf{I} - \mathbf{T}_{iso}\mathbf{A}^{12}\mathbf{T}_{sub}\mathbf{A}^{21}]^{-1}\mathbf{T}_{iso}$.

Execution Routine: By ingesting the serialized `excitation_metadata.json`, the script reconstructs the exact orthogonal incident polarization vectors. It constructs the scattered $[\mathbf{F}]$ and incident $[\mathbf{A}]$ coefficient matrices and rigorously extracts the invariant $\mathbf{T}$ operator utilizing the Moore-Penrose pseudoinverse.

2.3 Stage 3: Macroscopic Array Aggregation (Planned)

The final stage manages the multi-scale aggregation of the sparse scatterer ensembles. The localized unit-cell transition matrices ($\mathbf{T}^j$) are aggregated using the Foldy-Lax multiple scattering equations [3, 4]. The scattered VSWFs from each cell are analytically projected onto the local coordinate systems of neighboring cells via the translation addition theorems ($\mathbf{A}^{ij}$) [4]. The inversion of this global coupled block matrix yields the total macroscopic scattered field. This field is projected onto specific propagating plane waves to extract the standard network scattering parameters (S_{21}, S_{11}) for the full metasurface.

3 Theoretical Foundations

The following sections detail the exact theoretical formalisms governing the transition matrix evaluation for modeling and designing large-scale metasurfaces.

3.1 Vector Spherical Harmonics and Multi-Pole Expansion

The macroscopic scattering environment is modeled using the transition matrix method, which necessitates the expansion of incident and scattered fields into vector spherical harmonics [5, 6]. For spherical expansions, the generating scalar function ψ_{nm} solves the scalar Helmholtz equation in spherical coordinates and is expressed as a product of radial and angular dependencies [7]:

$$\psi_{nm}^{(c)}(r, \theta, \phi) = z_n^{(c)}(kr)P_n^m(\cos \theta)e^{jm\phi} \quad (1)$$

Substituting the scalar Debye potential ψ_{nm} into the vector wave formulation, and utilizing the radial position vector $\mathbf{r}$ as the pilot vector, we obtain the magnetic (TE) and electric (TM) vector spherical wave functions (VSWFs) [8]:

$$\mathbf{M}_{nm}^{(c)}(\mathbf{r}) = \nabla \times (\mathbf{r}\psi_{nm}^{(c)}(\mathbf{r})) \quad (2)$$

$$\mathbf{N}_{nm}^{(c)}(\mathbf{r}) = \frac{1}{k} \nabla \times \mathbf{M}_{nm}^{(c)}(\mathbf{r}) \quad (3)$$

Consequently, any arbitrary incident electric field $\mathbf{E}_{inc}(\mathbf{r})$ interacting with an environmental scatterer can be rigorously expanded as [5]:

$$\mathbf{E}_{inc}(\mathbf{r}) = \sum_{n=1}^{n_{max}} \sum_{m=-n}^n \left[a_{nm} \mathbf{M}_{nm}^{(1)}(\mathbf{r}) + b_{nm} \mathbf{N}_{nm}^{(1)}(\mathbf{r}) \right] \quad (4)$$

The resulting scattered field $\mathbf{E}_{sca}(\mathbf{r})$ is analogously expanded utilizing the outward-propagating basis functions $\mathbf{M}_{nm}^{(3)}$ and $\mathbf{N}_{nm}^{(3)}$:

$$\mathbf{E}_{sca}(\mathbf{r}) = \sum_{n=1}^{n_{max}} \sum_{m=-n}^n \left[f_{nm} \mathbf{M}_{nm}^{(3)}(\mathbf{r}) + g_{nm} \mathbf{N}_{nm}^{(3)}(\mathbf{r}) \right] \quad (5)$$

3.2 Multipole Truncation and the Size Parameter

The infinite VSWF series must be numerically truncated at a maximum order, L_{max} [6]. This limit is governed by the dimensionless size parameter $x_{max} = k_{max}a$. It is mathematically critical that a represents the radius of the minimum circumscribing sphere enclosing the physical metamaterial geometry. L_{max} is determined via Wiscombe's bounds [6]:

For $0.02 \leq x_{max} \leq 8$:

$$L_{max} = \lceil x_{max} + 4x_{max}^{1/3} + 1 \rceil \quad (6)$$

For $8 < x_{max} < 4200$:

$$L_{max} = \lceil x_{max} + 4.05x_{max}^{1/3} + 2 \rceil \quad (7)$$

For $x_{max} \geq 4200$:

$$L_{max} = \lceil x_{max} + 4x_{max}^{1/3} + 2 \rceil \quad (8)$$

Excluding the non-physical monopole, the total number of unknown VSWF coefficients for both electric and magnetic polarizations is defined by the dimension N [8]:

$$N = 2 \sum_{l=1}^{L_{max}} (2l+1) = 2L_{max}(L_{max}+2) \quad (9)$$

3.3 The Transition Matrix (T-Matrix) Method

The transition matrix encapsulates the complete electromagnetic scattering properties of an object [5]. To formulate the linear operator $\mathbf{T}$, the complex expansion coefficients are vertically concatenated into block column vectors. The input vector $\mathbf{a}$ and output vector $\mathbf{f}$ are defined as:

$$\mathbf{a} = \begin{bmatrix} a_{nm} \\ b_{nm} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_{nm} \\ g_{nm} \end{bmatrix} \quad (10)$$

The scattering process is thereby governed by the rigorous linear relation [9]:

$$\mathbf{f} = \mathbf{T}\mathbf{a} \quad (11)$$

Because the vectors $\mathbf{a}$ and $\mathbf{f}$ contain both TE and TM mode coefficients, the T-matrix operator $\mathbf{T}$ assumes a partitioned 2×2 block matrix structure [8]:

$$\mathbf{T} = \begin{bmatrix} \mathbf{T}^{11} & \mathbf{T}^{12} \\ \mathbf{T}^{21} & \mathbf{T}^{22} \end{bmatrix} \quad (12)$$

Because the VSWF basis inherently fulfills the Helmholtz equation, $\mathbf{T}$ depends exclusively on the particle's geometry, permittivity tensor, and the excitation frequency, rendering it strictly invariant to the incident field profile [8].

3.4 Multiple Scattering Theory and Array Aggregation

For an array comprised of N localized unit cells, the self-consistent local exciting field coefficients for the i -th unit cell, $\mathbf{a}^i$, are rigorously defined by the Foldy-Lax multiple scattering equations [3]:

$$\mathbf{a}^i = \mathbf{a}_{inc}^i + \sum_{j \neq i}^N \mathbf{A}^{ij} \mathbf{f}^j \quad (13)$$

Yielding a coupled algebraic system governing the macroscopic array [4]:

$$\mathbf{a}^i - \sum_{j \neq i}^N \mathbf{A}^{ij} \mathbf{T}^j \mathbf{a}^j = \mathbf{a}_{inc}^i \quad (14)$$

3.5 Extraction of S-Parameters from Macroscopic Scattering

Assuming the metasurface array is illuminated by an incident plane wave characterized by a wavevector $\mathbf{k}_{inc}$ and a polarization state $\hat{\mathbf{e}}_{inc}$, the complex forward transmission coefficient (S_{21}) is extracted utilizing the forward scattering theorem [6, 7]:

$$S_{21} = 1 + \frac{j2\pi}{kA \cos \theta_{inc}} \hat{\mathbf{e}}_{inc}^* \cdot \mathbf{S}(\mathbf{k}_{inc}) \quad (15)$$

Similarly, the reflection parameter S_{11} is determined by evaluating the scattering amplitude in the specular reflection direction, $\mathbf{k}_{ref}$ [7]:

$$S_{11} = \frac{j2\pi}{kA \cos \theta_{inc}} \hat{\mathbf{e}}_{ref}^* \cdot \mathbf{S}(\mathbf{k}_{ref}) \quad (16)$$

References

- [1] Y. Deng, S. Ren, K. Fan, J. M. Malof, and W. J. Padilla, “Neural-adjoint method for the inverse design of all-dielectric metasurfaces,” *Optics Express*, vol. 29, no. 5, p. 7526, 2021.
- [2] J. M. Malof, S. Ren, and W. J. Padilla, “Forward and inverse design of artificial electromagnetic materials,” in *Advances in Electromagnetics Empowered by Artificial Intelligence and Deep Learning*, pp. 345–370, IEEE, 2023.
- [3] L. L. Foldy, “The multiple scattering of waves. I. general theory of isotropic scattering by randomly distributed scatterers,” *Physical Review*, vol. 67, no. 3-4, pp. 107–119, 1945.
- [4] L. Tsang, J. A. Kong, and K.-H. Ding, *Scattering of electromagnetic waves: theories and applications*. John Wiley & Sons, 2001.
- [5] P. C. Waterman, “Matrix formulation of electromagnetic scattering,” *Proceedings of the IEEE*, vol. 53, no. 8, pp. 805–812, 1965.
- [6] M. I. Mishchenko, L. D. Travis, and A. A. Lacis, *Scattering, Absorption, and Emission of Light by Small Particles*. Cambridge University Press, 2002.
- [7] J. D. Jackson, *Classical Electrodynamics*. John Wiley & Sons, 3rd ed., 1999.
- [8] M. I. Mishchenko, L. D. Travis, and D. W. Mackowski, “T-matrix computations of light scattering by nonspherical particles: A review,” *Journal of Quantitative Spectroscopy and Radiative Transfer*, vol. 55, no. 5, pp. 535–575, 1996.
- [9] P. C. Waterman, “Symmetry, unitarity, and geometry in electromagnetic scattering,” *Physical Review D*, vol. 3, no. 4, p. 825, 1971.